

How Do Hormones Control the Human Body

Definition: Hormone is a chemical substance produced in minute quantities by an endocrine gland. It is transported in the bloodstream to target organs where it exerts its effect(s).

#BiologyDefinitions

Hormone ⇒ can influence growth, development and activity of an organism.
⇒ they are chemical messengers that help the various parts of the body to respond, develop and work together smoothly.

after they have performed their functions, they are destroyed in the liver and excreted by the kidneys.

Since hormones' help to control and coordinate body activities ⇒ their production must be carefully controlled to prevent disastrous results.

(eg: person will suffer from diabetes if too little of the hormone insulin is produced in the body. Signs of diabetes include fatigue, weight loss, high blood glucose level and slow healing of wounds)

Where are Hormones Produced

⇒ produced by glands.

⇒ all glands we know about so far ⇒ have a duct or tube carrying away their secretions.

(eg: Salivary gland has salivary duct to carry saliva to the mouth)

Sweat gland ⇒ has a duct for carrying sweat out of the body.

such glands are called **Exocrine glands**

However ⇒ hormones are produced by **ductless glands**.

⇒ does not have duct to carry away its secretion

Hormone produced by ductless gland ⇒ secreted directly into bloodstream.

Bloodstream then distributed around around body

Ductless glands ⇒ known as **Endocrine Glands**

Some glands ⇒ purely endocrine glands, producing only hormones, eg pituitary gland that secretes ADH.

other glands such as pancreas ⇒ produce both hormones and other secretions.

- pancreas produce pancreatic juice, which is carried by the pancreatic duct to the duodenum
 - Pancreas also contains special group of cells known as *islets of Langerhans* which secrete hormones insulin and glucagon $\Rightarrow$ into bloodstream
- Hence $\Rightarrow$ pancreas can be considered both endocrine and exocrine gland

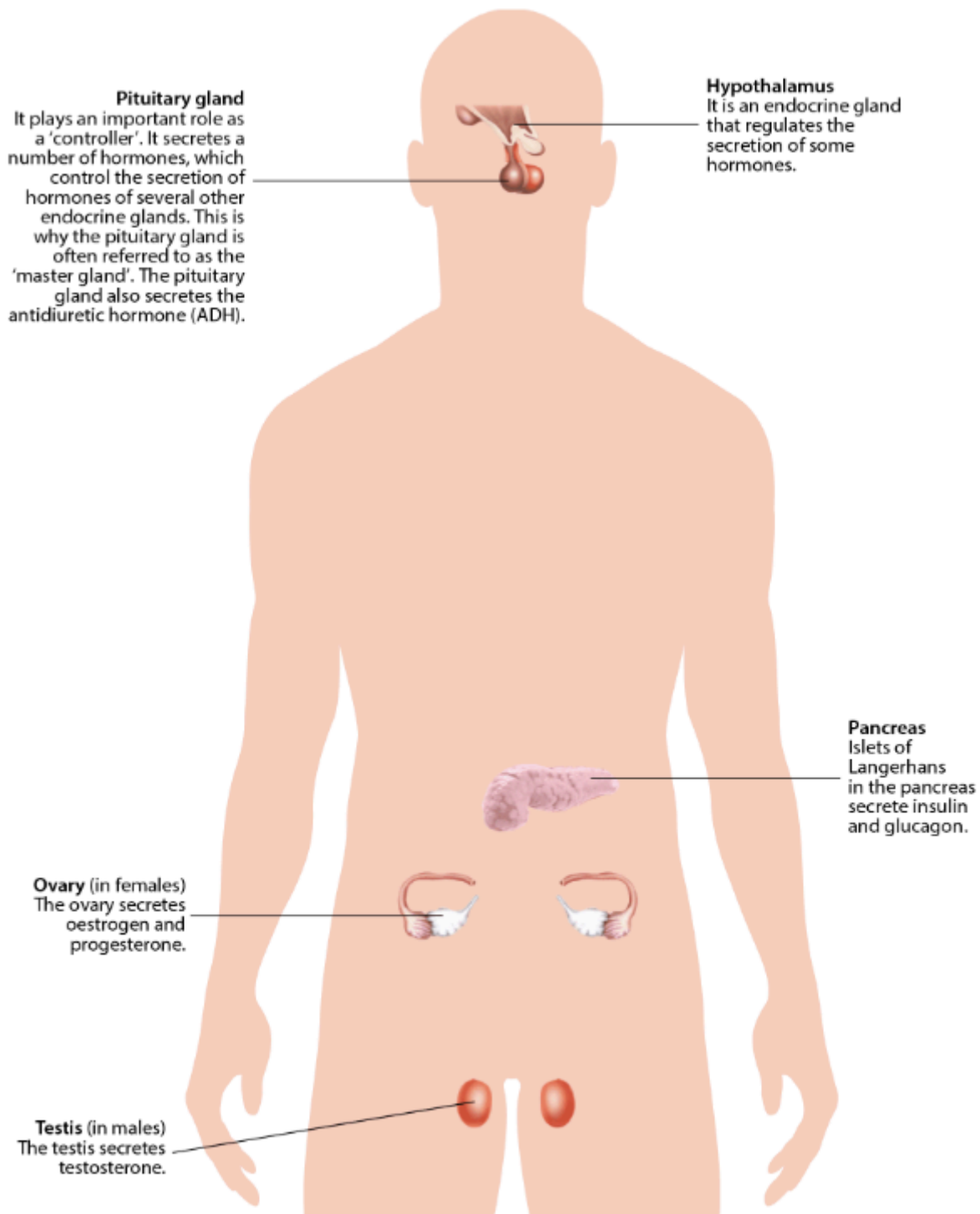


Figure 9.6 Location of some endocrine glands

Individual Effects of Insulin and Glucagon

[Effects of Insulin](#)

[Effects of Glucagon](#)
